Problem 1:

$\mathbb{P}(X=1) = \mathbb{P}(\{1,1\}) = \frac{1}{36}$

$\mathbb{P}(X=2) = \mathbb{P}(\{1,2\},\{2,1\}) = \frac{2}{36}$

$\mathbb{P}(X=3) = \mathbb{P}(\{1,3\},\{3,1\}) = \frac{2}{36}$

$\mathbb{P}(X=4) = \mathbb{P}(\{1,4\},\{4,1\},\{2,2\}) = \frac{3}{36}$

$\mathbb{P}(X=5) = \mathbb{P}(\{1,5\},\{5,1\}) = \frac{2}{36}$

$\mathbb{P}(X=6) = \mathbb{P}(\{1,6\},\{6,1\},\{2,3\},\{3,2\}) = \frac{4}{36}$

$\mathbb{P}(X=8) = \mathbb{P}(\{2,4\},\{4,2\}) = \frac{2}{36}$

$\mathbb{P}(X=9) = \mathbb{P}(\{3,3\}) = \frac{1}{36}$

$\mathbb{P}(X=10) = \mathbb{P}(\{2,5\},\{5,2\}) = \frac{2}{36}$

$\mathbb{P}(X=12) = \mathbb{P}(\{2,6\},\{6,2\},\{3,4\},\{4,3\}) = \frac{4}{36}$

$\mathbb{P}(X=15) = \mathbb{P}(\{3,5\},\{5,3\}) = \frac{2}{36}$

$\mathbb{P}(X=16) = \mathbb{P}(\{4,4\}) = \frac{1}{36}$

$\mathbb{P}(X=18) = \mathbb{P}(\{3,6\},\{6,3\}) = \frac{2}{36}$

$\mathbb{P}(X=20) = \mathbb{P}(\{4,5\},\{5,4\}) = \frac{2}{36}$

$\mathbb{P}(X=24) = \mathbb{P}(\{4,6\},\{6,4\}) = \frac{2}{36}$

$\mathbb{P}(X=25) = \mathbb{P}(\{5,5\}) = \frac{1}{36}$

$\mathbb{P}(X=30) = \mathbb{P}(\{5,6\},\{6,5\}) = \frac{2}{36}$

$\mathbb{P}(X=36) = \mathbb{P}(\{6,6\}) = \frac{1}{36}$

$$\therefore \mathbb{E}[X] = 1\cdot\tfrac{1}{36} + 2\cdot\tfrac{2}{36} + 3\cdot\tfrac{2}{36} + 4\cdot\tfrac{3}{36} + 5\cdot\tfrac{2}{36} + 6\cdot\tfrac{4}{36} + 8\cdot\tfrac{2}{36} + 9\cdot\tfrac{1}{36} + 10\cdot\tfrac{2}{36} + 12\cdot\tfrac{4}{36} + 15\cdot\tfrac{2}{36} + 16\cdot\tfrac{1}{36} + 18\cdot\tfrac{2}{36} + 20\cdot\tfrac{2}{36} + 24\cdot\tfrac{2}{36} + 25\cdot\tfrac{1}{36} + 30\cdot\tfrac{2}{36} + 36\cdot\tfrac{1}{36} = \tfrac{1}{36}\cdot 441$$

Problem 2: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\,dx$

$$= \int_0^{\infty} x f(x)\,dx + \int_{-\infty}^{0} x f(x)\,dx$$

$$= \int_0^{\infty} x f(x)\,dx$$

We integrate by part:

$u = x$ $\quad dv = f(x) = d(F_X(x)-1)$

$du = dx$ $\quad v = F_X(x) - 1$

$$\therefore \int_0^{\infty} x f(x)\,dx = \int_0^{\infty} x\,d(F_X(x)-1) = \int_0^{\infty} u\,dv = uv - \int_0^{\infty} v\,du$$

$$= [\,x(F_X(x)-1)]_0^{\infty} - \int_0^{\infty} (F_X(x)-1)\,dx$$

$$= [-x(1-F_X(x))]_0^{\infty} + \int_0^{\infty} \mathbb{P}(X \geq x)\,dx$$

$$= 0 + \int_0^{\infty} \mathbb{P}(X \geq x)\,dx$$

$$\therefore \mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X \geq t)\,dt$$